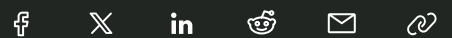
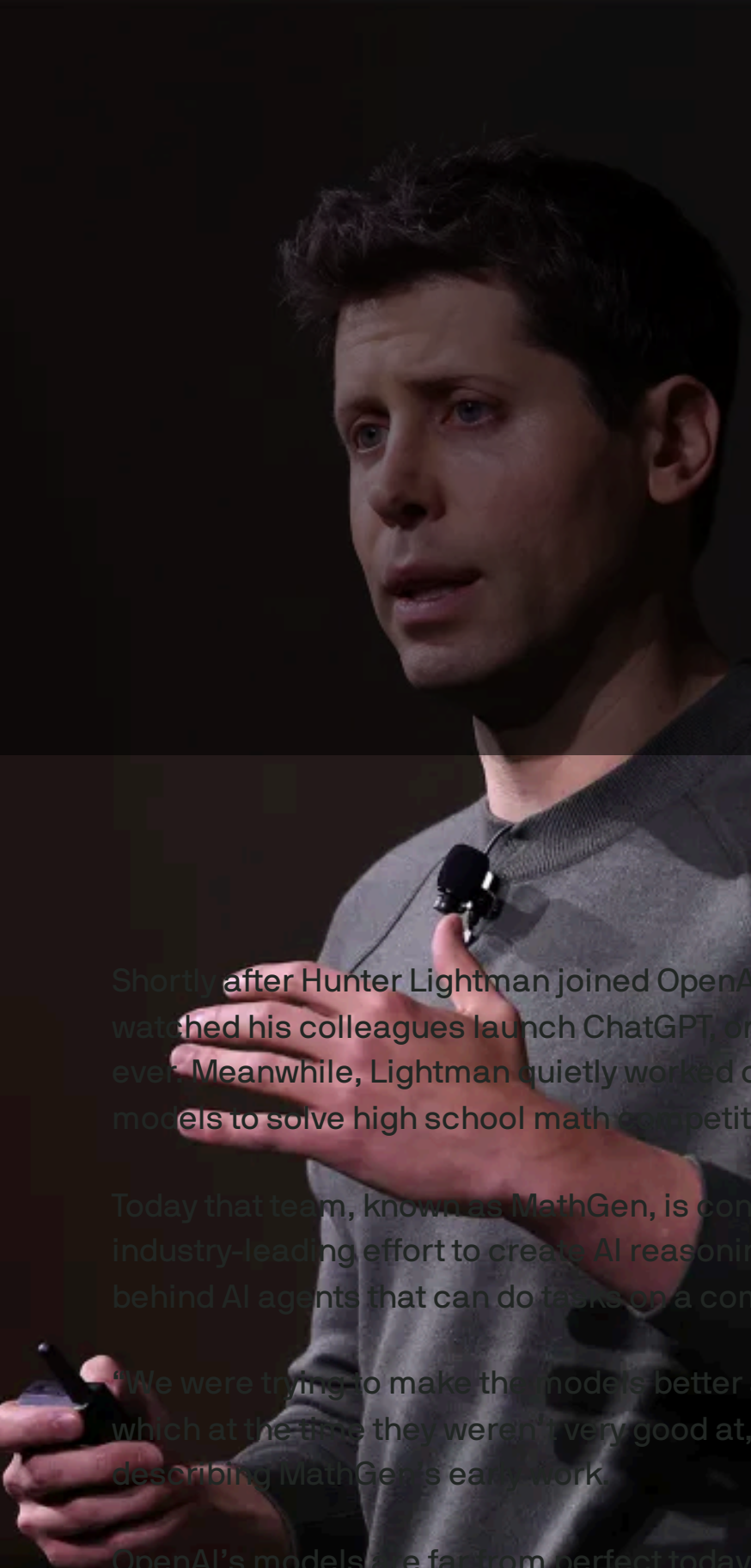


AI



Inside OpenAI's quest to  
make AI do anything for you



How OpenAI's first reasoning model sparked Silicon Valley's obsession with general-purpose AI agents — and what comes next.

Maxwell Zeff

7:00 AM PDT · August 3, 2025

Shortly after Hunter Lightman joined OpenAI as a researcher in 2022, he watched his colleagues launch ChatGPT, one of the fastest-growing products ever. Meanwhile, Lightman quietly worked on a team teaching OpenAI's models to solve high school math competitions.

Today that team, known as MathGen, is considered instrumental to OpenAI's industry-leading effort to create AI reasoning models: the core technology behind AI agents that can do tasks on a computer like a human would.

"We were trying to make the models better at mathematical reasoning, which at the time they weren't very good at," Lightman told TechCrunch, describing MathGen's early work.

OpenAI's models are far from perfect today — the company's latest AI systems still hallucinate and its agents struggle with complex tasks.

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But its state-of-the-art models have improved significantly on mathematical reasoning. One of OpenAI's models recently won a [gold medal](#) at the International Mathematical Olympiad (IMO), a math competition for the world's brightest high school students. OpenAI believes these reasoning capabilities will translate to other subjects, and ultimately power general-purpose agents that the company has always dreamed of building.

ChatGPT was a happy accident — a low-key research preview turned viral consumer business — but OpenAI's agents are the product of a yearslong, deliberate effort within the company.

“Eventually, you’ll just ask the computer for what you need and it’ll do all of these tasks for you,” said OpenAI CEO Sam Altman at the company’s [first developer conference](#) in 2023. “These capabilities are often talked about in the AI field as agents. The upsides of this are going to be tremendous.”



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OPENAI CEO SAM ALTMAN SPEAKS DURING THE OPENAI DEVDAY EVENT ON NOVEMBER 06, 2023, IN SAN FRANCISCO, CALIFORNIA.

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Whether agents will meet Altman's vision remains to be seen, but OpenAI shocked the world with the release of [its first AI reasoning model, o1](#), in the fall of 2024. Less than a year later, the 21 foundational researchers behind that breakthrough are the most highly sought-after talent in Silicon Valley.

Mark Zuckerberg [recruited](#) five of the o1 researchers to work on Meta's new superintelligence-focused unit, offering some compensation packages north of \$100 million. One of them, Shengjia Zhao, was recently [named](#) chief scientist of Meta Superintelligence Labs.

## The reinforcement learning renaissance

The rise of OpenAI's reasoning models and agents are tied to a machine learning training technique known as reinforcement learning (RL). RL provides feedback to an AI model on whether its choices were correct in simulated environments.

RL has been used for decades. For instance, in 2016, about a year after OpenAI was founded in 2015, an AI system created by Google DeepMind using RL, [AlphaGo](#), gained global attention after beating a world champion in the board game Go.



SOUTH KOREAN PROFESSIONAL GO PLAYER LEE SE-DOL (R) PREPARES FOR HIS FOURTH MATCH AGAINST GOOGLE'S ARTIFICIAL INTELLIGENCE PROGRAM, ALPHAGO, DURING THE GOOGLE DEEPMIND CHALLENGE MATCH ON MARCH 13, 2016, IN SEOUL, SOUTH KOREA.

IMAGE CREDITS:GOOGLE / GETTY IMAGES

Around that time, one of OpenAI's first employees, Andrej Karpathy, began pondering how to leverage RL to create an AI agent that could use a computer. But it would take years for OpenAI to develop the necessary models and training techniques.

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By 2018, OpenAI pioneered its first large language model in the GPT series, pretrained on massive amounts of internet data and large clusters of GPUs. GPT models excelled at text processing, eventually leading to ChatGPT, but struggled with basic math.

It took until 2023 for OpenAI to achieve a breakthrough, initially dubbed "Q\*" and then "Strawberry," by combining LLMs, RL, and a technique called "test-

time computation.” The latter gave the models extra time and computing power to plan and work through problems, verifying its steps, before providing an answer.

This allowed OpenAI to introduce a new approach called “chain-of-thought” (CoT), which improved AI’s performance on math questions the models hadn’t seen before.

“I could see the model starting to reason,” said OpenAI researcher Ahmed El-Kishky. “It would notice mistakes and backtrack, it would get frustrated. It really felt like reading the thoughts of a person.”

Though individually these techniques weren’t novel, OpenAI uniquely combined them to create Strawberry, which directly led to the development of o1. OpenAI quickly identified that the planning and fact-checking abilities of AI reasoning models could be useful to power AI agents.

“We had solved a problem that I had been banging my head against for a couple of years,” said Lightman. “It was one of the most exciting moments of my research career.”

## Scaling reasoning

With AI reasoning models, OpenAI determined it had two new axes that would allow it to improve AI models: using more computational power during the post-training of AI models, and giving AI models more time and processing power while answering a question.

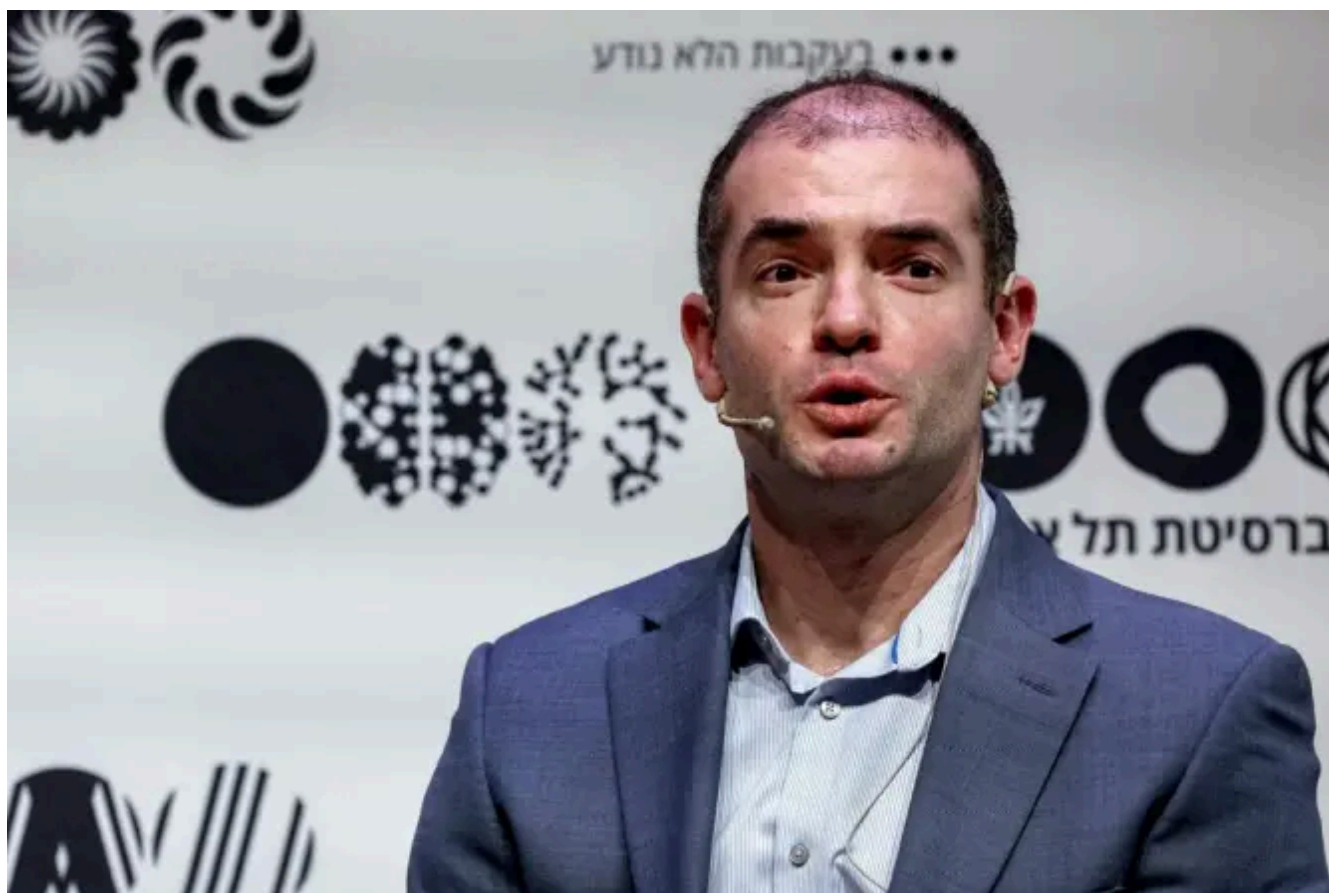
“OpenAI, as a company, thinks a lot about not just the way things are, but the way things are going to scale,” said Lightman.

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Shortly after the 2023 Strawberry breakthrough, OpenAI spun up an “Agents” team led by OpenAI researcher Daniel Selsam to make further progress on this new paradigm, two sources told TechCrunch. Although the team was called “Agents,” OpenAI didn’t initially differentiate between reasoning models and agents as we think of them today. The company just wanted to make AI systems capable of completing complex tasks.

Eventually, the work of Selsam’s Agents team became part of a larger project to develop the o1 reasoning model, with leaders including OpenAI co-founder Ilya Sutskever, chief research officer Mark Chen, and chief scientist Jakub Pachocki.



ILYA SUTSKEVER, COMPUTER SCIENTIST AND CO-FOUNDER AND FORMER CHIEF SCIENTIST OF OPENAI.  
IMAGE CREDITS: JACK GUEZ / AFP / GETTY IMAGES

OpenAI would have to divert precious resources — mainly talent and GPUs — to create o1. Throughout OpenAI’s history, researchers have had to negotiate with company leaders to obtain resources; demonstrating breakthroughs was a surefire way to secure them.

“One of the core components of OpenAI is that everything in research is bottom up,” said Lightman. “When we showed the evidence [for o1], the company was like, ‘This makes sense, let’s push on it.’”

Some former employees say that the startup’s mission to develop AGI was the key factor in achieving breakthroughs around AI reasoning models. By

focusing on developing the smartest-possible AI models, rather than products, OpenAI was able to prioritize o1 above other efforts. That type of large investment in ideas wasn't always possible at competing AI labs.

The decision to try new training methods proved prescient. By late 2024, several leading AI labs started seeing [diminishing returns](#) on models created through traditional pretraining scaling. Today, much of the AI field's momentum comes from advances in reasoning models.

## What does it mean for an AI to “reason”?

In many ways, the goal of AI research is to re-create human intelligence with computers. Since the launch of o1, ChatGPT's UX has been filled with more human-sounding features such as “thinking” and “reasoning.”

When asked whether OpenAI's models were truly reasoning, El-Kishky hedged, saying he thinks about the concept in terms of computer science.

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“We're teaching the model how to efficiently expend compute to get an answer. So if you define it that way, yes, it is reasoning,” said El-Kishky.

Lightman takes the approach of focusing on the model's results and not as much on the means or their relation to human brains.





IMAGE CREDITS:DEVIN COLDEWEY

“If the model is doing hard things, then it is doing whatever necessary approximation of reasoning it needs in order to do that,” said Lightman. “We can call it reasoning, because it looks like these reasoning traces, but it’s all just a proxy for trying to make AI tools that are really powerful and useful to a lot of people.”

OpenAI’s researchers note people may disagree with their nomenclature or definitions of reasoning — and surely, [critics have emerged](#) — but they argue it’s less important than the capabilities of their models. Other AI researchers tend to agree.

Nathan Lambert, an AI researcher with the non-profit AI2, compares AI reasoning modes to airplanes in a [blog post](#). Both, he says, are man-made systems inspired by nature — human reasoning and bird flight, respectively — but they operate through entirely different mechanisms. That doesn’t make them any less useful, or any less capable of achieving similar outcomes.

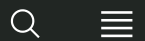
A group of AI researchers from OpenAI, Anthropic, and Google DeepMind agreed in a recent [position paper](#) that AI reasoning models are not well understood today, and more research is needed. It may be too early to confidently claim what exactly is going on inside them.

# The next frontier: AI agents for subjective tasks

The AI agents on the market today work best for well-defined, verifiable domains such as coding. OpenAI's [Codex agent](#) aims to help software engineers offload simple coding tasks. Meanwhile, Anthropic's models have become particularly [popular](#) in AI coding tools like Cursor and Claude Code — these are some of the first AI agents that people are willing to [pay up for](#).

However, general-purpose AI agents like OpenAI's [ChatGPT agent](#) and Perplexity's [Comet](#) struggle with many of the complex, subjective tasks people want to automate. When trying to use these tools for online shopping or finding a long-term parking spot, I've found the agents take longer than I'd like and make [silly mistakes](#).

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“Like many problems in machine learning, it’s a data problem,” said Lightman when asked about the limitations of agents on subjective tasks. “Some of the research I’m really excited about right now is figuring out how to train on less verifiable tasks. We have some leads on how to do these things.”

Noam Brown, an OpenAI researcher who helped create the IMO model and o1, told TechCrunch that OpenAI has new general-purpose RL techniques that allow them to teach AI models skills that aren’t easily verified. This was how the company built the model that achieved a gold medal at IMO, he said.

OpenAI’s IMO model was a newer AI system that spawns multiple agents, which then simultaneously explore several ideas and then choose the best possible answer. These types of AI models are becoming more popular; [Google](#) and [xAI](#) have recently released state-of-the-art models using this technique.

“I think these models will become more capable at math, and I think they’ll get more capable in other reasoning areas as well,” said Brown. “The progress has been incredibly fast. I don’t see any reason to think it will slow down.”

These techniques may help OpenAI's models become more performant, gains that could show up in the company's upcoming GPT-5 model. OpenAI hopes to assert its dominance over competitors with the launch of GPT-5, ideally offering the [best AI model](#) to power agents for developers and consumers.

But the company also wants to make its products simpler to use. El-Kishky says OpenAI wants to develop AI agents that intuitively understand what users want, without requiring them to select specific settings. He says OpenAI aims to build AI systems that understand when to call up certain tools, and how long to reason for.

These ideas paint a picture of an ultimate version of ChatGPT: an agent that can do anything on the internet for you, and understand how you want it to be done. That's a much different product than what ChatGPT is today, but the company's research is squarely headed in this direction.

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While OpenAI undoubtedly led the AI industry a few years ago, the company now faces a tranche of worthy opponents. The question is no longer just whether OpenAI can deliver its agentic future, but can the company do so before Google, Anthropic, xAI, or Meta beat them to it?

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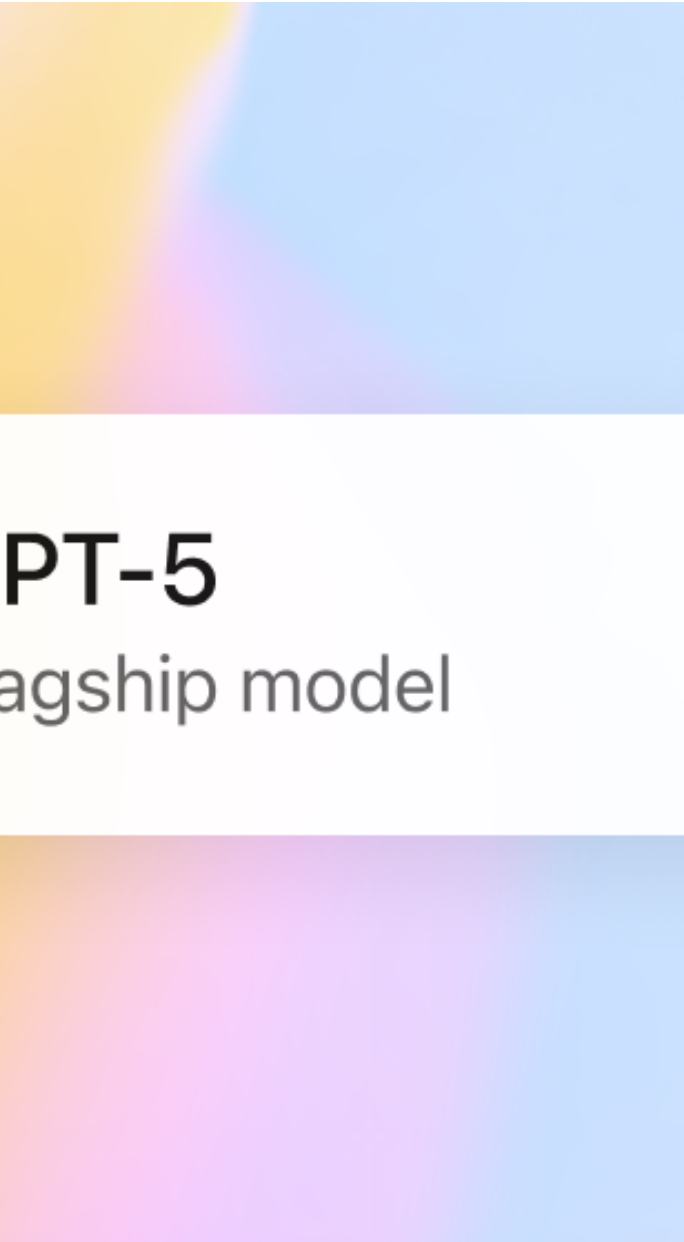
Maxwell Zeff

Senior AI Reporter | X

Maxwell Zeff is a senior reporter at TechCrunch specializing in AI. Previously with Gizmodo, Bloomberg, and MSNBC, Zeff has covered the rise of AI and the Silicon Valley Bank crisis. He is based in San Francisco. Wh...

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GPT-5  
flagship model

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OpenAI's GPT-5 is here

Maxwell Zeff — 10:00 AM PDT · August 7, 2025

OpenAI has launched GPT-5, a new flagship AI model that will power the company’s next generation of ChatGPT.

GPT-5, which was released Thursday, is OpenAI’s first “unified” AI model and combines the reasoning abilities of its [o-series](#) of models with the fast responses of its [GPT series](#). The next-generation model signals a new era for ChatGPT — and its creator, OpenAI — pointing to OpenAI’s broader ambitions to develop AI systems that are [more like agents](#) than chatbots.

While [GPT-4](#) enabled AI chatbots to offer smart responses on a wide variety of questions, GPT-5 allows ChatGPT to complete a wide variety of tasks on behalf of users — such as generating software applications, navigating a user’s calendar, or creating research briefs.

With GPT-5, OpenAI has also sought to make ChatGPT simpler to use. Instead of asking users to choose the right settings, GPT-5 comes equipped with a real-time router that decides how to offer the best answer, whether that’s responding to user questions quickly or taking additional time to “think” through answers.

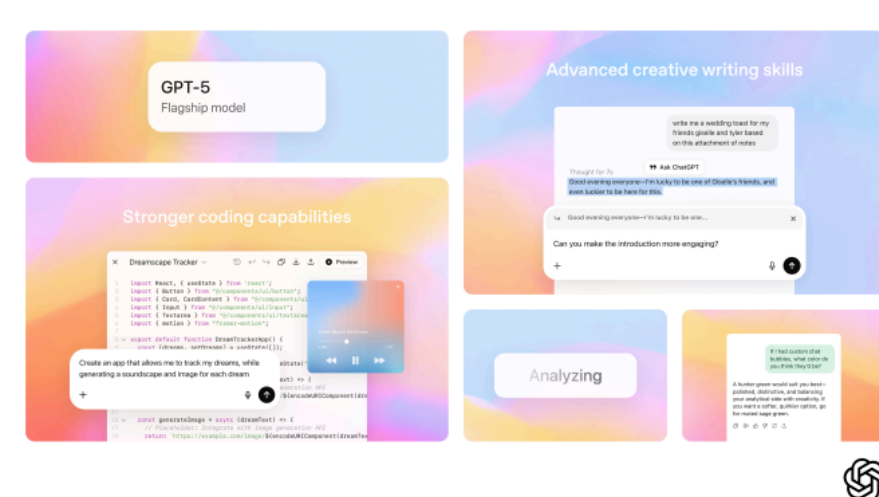


IMAGE CREDITS: OPENAI

During a briefing with reporters, OpenAI CEO Sam Altman claimed GPT-5 is “the best model in the world,” and said it represented a “significant step”



along the company’s path to developing AI that can outperform humans at most economically valuable work — that is, artificial general intelligence (AGI).

“Having something like GPT-5 would be pretty much unimaginable at any previous time in history,” said Altman.

Starting Thursday, GPT-5 will be available to all free users of ChatGPT as their default model. OpenAI’s VP of ChatGPT, Nick Turley, said this is part of the company’s effort to give free users access to an AI reasoning model for the first time. (Previously, the company gated these more advanced models behind a paywall.)

“This is just one of the ways that I’m excited to live the mission, making sure that this stuff actually benefits people,” said Turley on the decision, referencing OpenAI’s long-standing [mission](#) to distribute advanced AI to as many people as possible.



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The expectations are high for GPT-5, one of OpenAI's most anticipated product launches since ChatGPT put the company on the map in 2022. Since then, ChatGPT has grown into one of the world's most popular consumer products, reaching more than 700 million users every week — nearly 10% of the globe's population, according to the company.

Many see GPT-5 as a bellwether for AI progress broadly, and the model's reception by Silicon Valley could have profound implications for Big Tech, Wall Street, and policymakers regulating technology. These stakeholders are watching to see if GPT-5 offers a significant jump in AI's capabilities, much like its predecessor, GPT-4, which challenged expectations of what software can do.

## GPT-5 offers a slight edge on the competition

OpenAI claims GPT-5 is state-of-the-art in several domains, slightly edging out leading AI models from Anthropic, Google DeepMind, and Elon Musk's xAI on key benchmarks. However, GPT-5 slightly underperforms frontier AI models in other areas.

The company says GPT-5 offers frontier-level performance around coding; Altman said the model specifically excels at spinning up entire software applications on demand, in what's become known as "vibe coding."

On SWE-bench Verified — a test of real-world coding tasks pulled from GitHub — GPT-5 scores 74.9% on its first attempt. That means GPT-5 just outperforms Anthropic's latest Claude Opus 4.1 model, which scored [74.5%](#), and Google DeepMind's Gemini 2.5 Pro, which scored [59.6%](#).

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On Humanity's Last Exam — a difficult test measuring AI model performance across math, humanities, and the natural sciences — a version of GPT-5 with extended reasoning (GPT-5 Pro) scored 42% when using tools. That's slightly less than xAI was able to achieve with [Grok 4 Heavy](#), which scored 44.4% on the test.

IMAGE CREDITS: OPENAI

On GPQA Diamond — a test of PhD-level science questions — GPT-5 pro scored 89.4% on its first try, outperforming Claude Opus 4.1, which scored 80.9%, and Grok 4 Heavy, which scored 88.9%.

OpenAI says GPT-5 is better for answering health-related questions. On a test measuring accuracy in AI model responses around healthcare topics, HealthBench Hard Hallucinations, OpenAI says GPT-5 (with thinking) hallucinates just 1.6% of the time. This is far lower than the company's previous GPT-4o and o3 models, which scored 12.9% and 15.8, respectively.

While AI chatbots are not medical professionals, [millions of people](#) are using them for health advice. In response to this phenomenon, the company says GPT-5 is more proactive about flagging potential health concerns and helping users parse medical results.

In addition, OpenAI says GPT-5 is better than other AI models on more difficult-to-measure, [subjective domains](#), such as creative design and writing.

Turley said GPT-5 responds more naturally and exhibits “better taste” than other AI models on creative tasks.

“The vibes of this model are really good,” said Turley.

GPT-5 is also more accurate than OpenAI’s previous models, and the company says it suffers far less from hallucinations — the tendency for AI models to make up information — compared to its o-series models. Hallucinations seemed to be [getting worse](#) in OpenAI’s latest AI reasoning models, such as o3, and OpenAI previously said it didn’t quite understand why it was happening.

In responses to ChatGPT prompts, OpenAI found that GPT-5 (with thinking) hallucinates and responds with incorrect information 4.8% of the time. That’s a significant reduction from o3 and GPT-4o, which score hallucination rates of 22% and 20.6%, respectively, on the test.

On a benchmark measuring an AI model’s agentic ability to complete simulated online tasks, Tau-bench, GPT-5 offers mixed performance. On part of the test measuring an AI’s ability to navigate an airline’s website, GPT-5 scores 63.5%, slightly underperforming o3, which scored 64.8%. On another part of the test measuring AI’s ability to navigate retail websites, GPT-5 scores 81.1%, underperforming Claude Opus 4.1, which scored 82.4%.

OpenAI also says that GPT-5 is safer than its previous models. While AI reasoning models occasionally exhibit a tendency to [scheme against humans](#) or lie to promote their own goals, OpenAI

found that GPT-5 was deceptive at a lower rate than other models.

Alex Beutel, OpenAI’s safety research lead, said reducing deception improves not only the safety of GPT-5, but also the user experience, creating a model that’s more “transparent and honest in ways users can trust.”

Beutel also notes GPT-5 is better at discerning between bad actors who are trying to misuse ChatGPT and users making harmless requests. This results in GPT-5 being able to refuse more unsafe questions, while offering fewer rejections to users seeking harmless information.

## Upgrades for consumers and developers

ChatGPT is getting a few user experience upgrades as part of the GPT-5 launch. Users can now select from four new personalities in ChatGPT’s setting: Cynic, Robot, Listener, and Nerd. The company says these will adapt ChatGPT’s responses without requiring users to specifically ask the model to respond in a certain way.

Subscribers to ChatGPT’s \$20-per-month Plus plan get higher usage limits for GPT-5 than free users. Meanwhile, \$200-per-month Pro subscribers will have unlimited access to GPT-5, as well as a souped-up version called GPT-5 Pro that uses additional computational resources to produce better answers. Organizations on OpenAI’s Team, Edu, and Enterprise plans will gain access to GPT-5 as their default model next week.

For developers, GPT-5 is coming to OpenAI’s API in three sizes — gpt-5, gpt-5-mini, and gpt-5-nano — which will spend more or less time “reasoning” through tasks. Developers can also now control

verbosity in the OpenAI API, deciding how long or short an AI model's responses should be.

The base model of GPT-5 will cost developers \$1.25 per million input tokens (roughly 750,000 words, longer than the entire "Lord of the Rings" series) and \$10 per million output tokens.

The launch of GPT-5 comes after a busy week for OpenAI. The company released an open-weight reasoning model, [gpt-oss](#), that developers and enterprises can download for free and run at a fraction of the cost. The open model nearly matched the abilities of OpenAI's previous top models, o3 and o4-mini, but GPT-5 sets a new standard for frontier performance in some areas, such as coding.

However, GPT-5 seems to be roughly on par with other frontier AI models in several areas. Benchmarks, of course, only tell part of the story for any AI model, and it remains to be seen how developers will use GPT-5 in the real world, and whether the model is truly a step above the competition.

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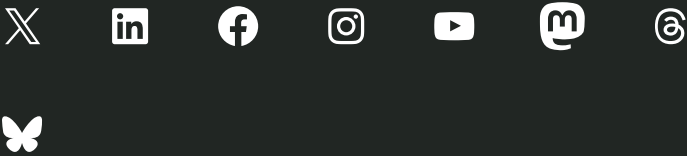
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